

PHYS 230 - Electricity and Magnetism

Lecture 15

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

3 Electric Potential Energy and Potential

- Differentials and Gradient

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

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Example:

$$d(3x^2) = \frac{d(3x^2)}{dx} dx$$

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Differential $df(x)$ of a function $f(x)$: an infinitesimal change in the dependent variable f , when the dependent variable changes by an infinitesimal amount dx

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Example:

$$d(3x^2) = \underbrace{\frac{d(3x^2)}{dx}}_{6x} dx$$

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Example:

$$d(3x^2) = \underbrace{\frac{d(3x^2)}{dx}}_{6x} dx = 6x dx$$

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$$df(x) = \frac{df(x)}{dx} dx$$

This can also be written as inner product of two vectors

$$df(x) = \left(\frac{df(x)}{dx} \hat{\mathbf{i}} \right) \cdot (dx \hat{\mathbf{i}})$$

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Differentials And Gradient

For a function of several variables $f(x, y, z)$ we have

$$df = \frac{\partial f(x, y, z)}{\partial x} dx + \frac{\partial f(x, y, z)}{\partial y} dy + \frac{\partial f(x, y, z)}{\partial z} dz$$

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Example:

$$d(3x^2 + 4y + \ln(z))$$

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Example:

$$d(3x^2 + 4y + \ln(z)) = \underbrace{\frac{\partial(3x^2 + 4y + \ln(z))}{\partial x}}_{6x} dx + \underbrace{\frac{\partial(3x^2 + 4y + \ln(z))}{\partial y}}_4 dy + \underbrace{\frac{\partial(3x^2 + 4y + \ln(z))}{\partial z}}_{\frac{1}{z}} dz$$

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Example:

$$\begin{aligned} d(3x^2 + 4y + \ln(z)) &= \underbrace{\frac{\partial (3x^2 + 4y + \ln(z))}{\partial x}}_{6x} dx + \underbrace{\frac{\partial (3x^2 + 4y + \ln(z))}{\partial y}}_4 dy + \underbrace{\frac{\partial (3x^2 + 4y + \ln(z))}{\partial z}}_{\frac{1}{z}} dz \\ &= 6x dx + 4dy + \frac{1}{z} dz \end{aligned}$$

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Differentials And Gradient

So we learned in Cartesian coordinates, **gradient** (2D or 3D derivative) is

$$\nabla = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z}$$

Differentials And Gradient

So we learned in Cartesian coordinates, **gradient** (2D or 3D derivative) is

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and **differential** is

$$df = \nabla f \cdot d\mathbf{r}$$

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and **differential** is

$$df = \nabla f \cdot d\mathbf{r}$$

which means that in 3D we have

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

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and **differential** is

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which means that in 3D we have

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

which in 1D (when $f = f(x)$ only function of 1 dimension x), becomes

$$df(x) = \underbrace{\frac{\partial f}{\partial x}}_{\frac{df}{dx}} dx + \underbrace{\frac{\partial f}{\partial y}}_0 dy + \underbrace{\frac{\partial f}{\partial z}}_0 dz \Rightarrow df = \frac{df}{dx} dx$$

Differentials And Gradient

Remember, **gradient of a function is a vector!**

$$\nabla f = \hat{\mathbf{i}} \frac{\partial f}{\partial x} + \hat{\mathbf{j}} \frac{\partial f}{\partial y} + \hat{\mathbf{k}} \frac{\partial f}{\partial z}$$

Differentials And Gradient

Remember, **gradient of a function is a vector!**

$$\nabla f = \hat{\mathbf{i}} \frac{\partial f}{\partial x} + \hat{\mathbf{j}} \frac{\partial f}{\partial y} + \hat{\mathbf{k}} \frac{\partial f}{\partial z}$$

Gradient itself is **a vector operator**

$$\nabla = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z}$$

Differentials And Gradient

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Gradient itself is **a vector operator**

$$\nabla = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z}$$

it acts on (operates on = takes the derivative of) functions

Gradient of Electric Potential: Electric Field

Remember

$$dV = \frac{dU}{q_0}$$

Gradient of Electric Potential: Electric Field

Remember

$$dV = \frac{dU}{q_0} = \frac{-dW}{q_0}$$

Gradient of Electric Potential: Electric Field

Remember

$$dV = \frac{dU}{q_0} = \frac{-dW}{q_0} = \frac{-\mathbf{F} \cdot d\mathbf{r}}{q_0}$$

Gradient of Electric Potential: Electric Field

Remember

$$dV = \frac{dU}{q_0} = \frac{-dW}{q_0} = \frac{-\mathbf{F} \cdot d\mathbf{r}}{q_0} = \frac{-q_0 \mathbf{E} \cdot d\mathbf{r}}{q_0}$$

Gradient of Electric Potential: Electric Field

Remember

$$dV = \frac{dU}{q_0} = \frac{-dW}{q_0} = \frac{-\mathbf{F} \cdot d\mathbf{r}}{q_0} = \frac{-q_0 \mathbf{E} \cdot d\mathbf{r}}{q_0} = -\mathbf{E} \cdot d\mathbf{r}$$

Gradient of Electric Potential: Electric Field

Remember

$$dV = \frac{dU}{q_0} = \frac{-dW}{q_0} = \frac{-\mathbf{F} \cdot d\mathbf{r}}{q_0} = \frac{-q_0 \mathbf{E} \cdot d\mathbf{r}}{q_0} = -\mathbf{E} \cdot d\mathbf{r}$$

So

$$dV = -\mathbf{E} \cdot d\mathbf{r}$$

Gradient of Electric Potential: Electric Field

Remember

$$dV = \frac{dU}{q_0} = \frac{-dW}{q_0} = \frac{-\mathbf{F} \cdot d\mathbf{r}}{q_0} = \frac{-q_0 \mathbf{E} \cdot d\mathbf{r}}{q_0} = -\mathbf{E} \cdot d\mathbf{r}$$

So

$$dV = -\mathbf{E} \cdot d\mathbf{r}$$

But we learned that

$$dV = \nabla V \cdot d\mathbf{r}$$

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$$\mathbf{E} = -\nabla V \Rightarrow E_x \hat{\mathbf{i}} + E_y \hat{\mathbf{j}} + E_z \hat{\mathbf{k}} = - \left(\frac{\partial V}{\partial x} \hat{\mathbf{i}} + \frac{\partial V}{\partial y} \hat{\mathbf{j}} + \frac{\partial V}{\partial z} \hat{\mathbf{k}} \right)$$

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Gradient of Electric Potential: Summary

Summarizing

$$\mathbf{E} = -\nabla V$$

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Inner product with $d\mathbf{r}$ yields

$$\mathbf{E} \cdot d\mathbf{r} = -\nabla V \cdot d\mathbf{r}$$

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$$\mathbf{E} \cdot d\mathbf{r} = -\underbrace{\nabla V \cdot d\mathbf{r}}_{dV} \Rightarrow \boxed{dV = -\mathbf{E} \cdot d\mathbf{r}} \Rightarrow \boxed{dV = -E_x dx - E_y dy - E_z dz}$$

Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

For a ring of charge with radius a and total charge Q , the potential at a point P on the ring's symmetry axis a distance x from the center is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + a^2}}$$

Find the electric field vector at P .

Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

Consider the potential $V(x, y, z) = A(x^2 + 3y^2 - \ln(z))$ where A is a constant. Find the x and z components, E_x and E_z , of the electric field.

Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

Gradient Operator in Spherical Coordinates

Sometimes in spherically symmetric cases, it is much better to use the spherical form of the gradient operator

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\boldsymbol{\varphi}} \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \varphi}$$

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This means

$$\nabla f = \left(\frac{\partial f}{\partial r} \right) \hat{\mathbf{r}} + \left(\frac{1}{r} \frac{\partial f}{\partial \theta} \right) \hat{\boldsymbol{\theta}} + \left(\frac{1}{r \sin(\theta)} \frac{\partial f}{\partial \varphi} \right) \hat{\boldsymbol{\varphi}}$$

Exercise 3

<https://epoll.srv.ualberta.ca> – Code: MBD

Given that the potential of a point charge q at a distance r from it is $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$, find the electric field vector of the point charge